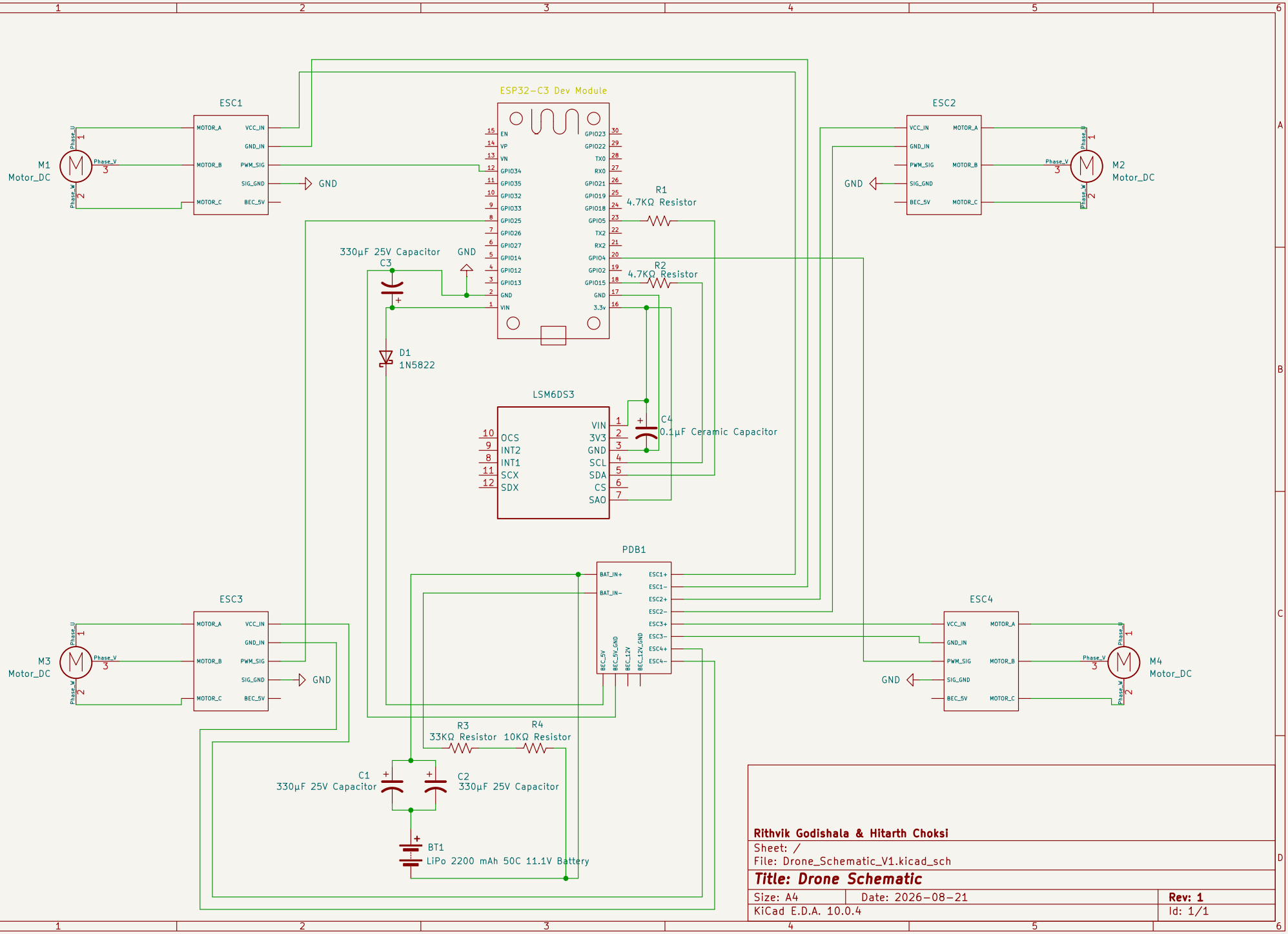

ESC1
MOTOR_A
VCC_IN
GND_IN
MOTOR_B
PWM_SIG
SIG_GND
MOTOR_C
BEC_5V
Phase_U
Phase_V
Phase_W
M1
Motor_DC
GND
ESP32-C3 Dev Module
EN
VP
VN
GPIO34
GPIO35
GPIO32
GPIO33
GPIO25
GPIO26
GPIO27
GPIO14
GPIO12
GPIO13
GND
VIN
GPIO23
GPIO22
TX0
RX0
GPIO21
GPIO19
GPIO18
GPIO5
TX2
RX2
GPIO4
GPIO2
GPIO15
GND
3.3v
R1
4.7KΩ Resistor
R2
4.7KΩ Resistor
330µF 25V Capacitor
C3
D1
1N5822
LSM6DS3
OCS
INT2
INT1
SCX
SDX
VIN
3V3
GND
SCL
SDA
CS
SAO
C4
0.1µF Ceramic Capacitor
ESC2
M2
Motor_DC
PDB1
BAT_IN+
BAT_IN-
ESC1+
ESC1-
ESC2+
ESC2-
ESC3+
ESC3-
ESC4+
ESC4-
BEC_5V
BEC_5V_GND
BEC_12V
BEC_12V_GND
ESC3
M3
Motor_DC
ESC4
M4
Motor_DC
R3
33KΩ Resistor
R4
10KΩ Resistor
C1
330µF 25V Capacitor
C2
330µF 25V Capacitor
BT1
LiPo 2200 mAh 50C 11.1V Battery
Rithvik Godishala & Hitarth Choksi
Sheet: /
File: Drone_Schematic_V1.kicad_sch
Title: Drone Schematic
Size: A4
Date: 2026-08-21
Rev: 1
KiCad E.D.A. 10.0.4
Id: 1/1